AI OPEN
DATA CENTER:
Power Distribution



Closing the Standards Gap for LVDC Power Distribution

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Understanding the Industry Need

- AI is driving power demand increases
  - Power is the main constraint
  - New power architectures are being developed
- Power Distribution Sub-Project launched at OCP Global Summit 2025
- UL Solutions was asked to lead the Codes & Standards workstream
- Safety is critical pathway for new power architectures

Power Distribution Sub-Project Workstreams

System Architecture Grid Topologies Redundancy Levels	**Voltage Levels and Power Quality** Nominal Voltages and Voltage Bands Ripple, Tolerance, and Stability	**Polarities & System Earthing** Polarities / Earthing systems Galvanic Isolation Corrosion Prevention
Protection and Safety Short-Circuit, Overload, Electric Shock Arc-Fault Protection Selectivity and Fault Isolation	**Main Components (high level requirements)** Supply (Rectifiers, AICs, SSTs) Distribution (Switchgears & switchboards) Protection (Overcurrent protection devices) Loads (server racks) * Energy Storage Systems **	**Codes and Standards** Status quo & Gap analysis Proposed gap filling actions
Control & Operation Sequence of Operation (SOO) Control Requirements Surrounding SOO	**EMC** EMC Applicability at product and system level Emissions, Immunity, Ripple Voltage and Harmonics Criteria Cable Shielding & Filtering	**Joint Workstream** **DCF PD + R&P**

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Workstream Goals

1. Industry alignment on requirements
2. Empowering market to confirm compliance
   - ✓ Testing – ensuring available capacity in the market
   - ✓ Labeling – creating the opportunity to label products
   - ✓ Training – electrical contractors, AHJs, etc.
3. Bringing organizations together



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Data center power – products covered (AC & DC)

Source
- Microgrids
- Battery energy storage systems
- Energy storage – stationary
- Medium voltage solid state power units
- Generators

Distribution equipment
- Switchgear
- Automatic transfer switches
- Remote power panels
- Busways and busbars
- Switchboards

Power supplies/ converters
- UPS
- Power supplies
- DC/DC converter

Overcurrent/ protection devices
- Circuit breakers and fuses
- Surge protection devices
- AFCI/GFCI
- Isolation monitors
- Ground fault sensing and relaying equipment



Datacenter Power Distribution

Switching/ control devices
- Disconnects
- Contactors
- Control panels

Grounding/ bonding
- Grounding/bonding equipment

Connectors
- Terminal blocks
- Bus plug connectors
- Pin and sleeve connectors
- Multi-point power cable assemblies
- Wiring connectors
- Power distribution blocks

Cables/cords
- AC/DC power whips
- Power cables
- Liquid-cooling cables
- Superconducting cable systems
- Extruded busbars

Power utilization
- Compute rack (GPUs, shelves, servers)

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Workstream Team Deliverables

- Whitepaper technical guidance
  - System-level standards
  - Product-level standards
  - Installation codes
  - Interim guidance
- Standards inventory across SDOs
  - UL Solutions, IEC, IEEE, NFPA, NEMA, NEC, Current/OS, ODCA,
- Priority Product Standards
  - Sidecars, modular, MV SST, circuit breakers, liquid-cooling





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US Safety Standards for power—800V DC data center

Source		Distribution equipment	
UL 3001	AC microgrids	**UL 1558**	Low-voltage switchgear
UL 3001D OOI	DC microgrids	**UL 1008**	Automatic transfer switches (ATS)
UL 9540	Battery energy storage systems (BESS)	**UL 67**	Remote power panel, panelboard, rack-mounted panelboard
UL 1973	Energy storage — stationary applications	**UL 857**	800 VDC busways and tap cans
UL 2877 OOI	Medium-voltage solid-state transformers (MV SST)	**UL 508**	Busbars
UL 2200	Engine Generator Assemblies	**UL 2140 OOI**	Prefabricated power distribution assemblies
		UL 2755 OOI	Prefabricated modular data center systems
		UL 891	DC switchboards
		UL 1400-1 OOI, UL 1400-2	Industrial fault managed power (FMP)
		UL 62368-1 + CRD	DC sidecars – 800V DC output
		UL 1008B	Source Interconnection Switches

Switching/control devices		Power supplies/converters	
UL 98, UL 98A OOI	DC disconnect	**UL 1012 + UL 1778**	DC UPS
UL 60947-4-1, UL 61810-1, UL 508	DC contactor	**UL 1012**	Power supplies
UL 6200	Controllers for Power Production	**UL 1741**	DC/DC converters
UL 508A	Industrial Control Panels		

Overcurrent/protection devices		Connectors	
UL 489	Molded-case circuit breakers and associated devices	**UL 1059, UL 60947-7-1**	Terminal blocks
UL 1066	Low-voltage power circuit breakers and associated devices	**UL 1977**	Busbar connectors
UL 489I	Solid-state circuit breakers	**UL 1691, UL 1682**	Single-pole, Pin & sleeve connectors
UL 248-4, -8, -10, -12, -15	Branch-circuit fuses (Class CC, J, L, R, T)	**UL 2237**	Multi-point interconnection power cable assemblies
UL 248-13	Semiconductor fuses	**New OOI**	Litz wire PDU2 & J-box connectors
UL 248-14	Supplemental fuses	**UL 486A, UL 486B**	Wire connectors
UL 1449	Surge protection devices	**UL 1953 OOI**	Power distribution blocks
New OOI	DC AFCI	**Grounding/bonding**	
New OOI	DC GFCI	**UL 467**	Grounding/bonding equipment
New OOI	Isolation monitors	**Power utilization**	
UL 1053 and New OOI	Ground-fault sensing and relaying equipment	**UL 62368-1**	Compute rack (GPUs, shelves, servers)
Cables/cords			
UL 817A OOI	AC/DC power whips	**New OOI**	Superconducting Cable Systems
UL 2137 OOI	Data center power cables	**New OOI**	Extruded Insulated Busbars
New OOI	Data center liquid-cooling cables	**New OOI**	Extruded Liquid Cooled Insulated Busbars
Reference standards			
UL 840	Insulation coordination including clearances and creepage distances	**UL RP 2986**	Measuring incident energy exposures

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Global/IEC Safety Standards for power—800V DC data

Source			Distribution equipment		
UL 3001	**IEC 60364 series + country specific requirements**	AC microgrids	UL 1558	**IEC 61439-2**	Low-voltage switchgear
UL 3001D OOI	**IEC 60364 series + country specific requirements**	DC microgrids	UL 1008	**IEC 60947-6-1**	Automatic transfer switches (ATS)
UL 9540	**IEC 62933 series + other applicable/overlapping standards**	Battery energy storage systems (BESS)	UL 67	**IEC 61439-2, -3**	Remote power panel, panelboard, rack-mounted panelboard
UL 1973	**IEC 62619, IEC 63056**	Energy storage — stationary applications	UL 857	**IEC 61439-6**	800 VDC busways and tap cans
UL 2877 OOI	**IEC 62477-2**	Medium-voltage solid-state transformers (MV SST)	UL 508	**IEC 61439-2**	Power distribution — DC busbars
UL 2200	**IEC 60034 series + many additional requirements**	Engine Generator Assemblies	UL 2140 OOI	**IEC 61439-2**	Prefabricated power distribution assemblies
			UL 2755 OOI	**ISO/IEC 22237 series**	Prefabricated modular data center systems
			UL 891	**IEC 61439-2**	DC switchboards
			UL 1400-1 OOI, UL 1400-2		Industrial fault managed power (FMP)
			UL 62368-1 + CRD	**IEC 62368-1, IEC 61439-2**	DC sidecars – 800V DC output
			UL 1008B	**EN IEC 62991:2025**	Source Interconnection switches

Switching/control devices			Power supplies/converters		
UL 98, UL 98A OOI	**IEC 60947-3**	DC disconnect	UL 1012 + UL 1778	**IEC 62040-5-1 IEC 62040-1, IEC 62477-1**	DC UPS
UL 60947-4-1, UL 61810-1, UL 508	**IEC 60947-4-1, IEC 61810-1, IEC 61810-10**	DC contactor	UL 1012	**IEC 62477-1 + appropriate part 2**	Power supplies
UL 6200		Controllers for Power Production	UL 1741	**IEC 62109 series**	DC/DC converters
UL 508A	**IEC 60204-1 (partly aligned with NFPA79)**	Industrial Control Panels			

IEC Mapped Standards (1 of 2)

Global/IEC Safety Standards for power—800V DC data center

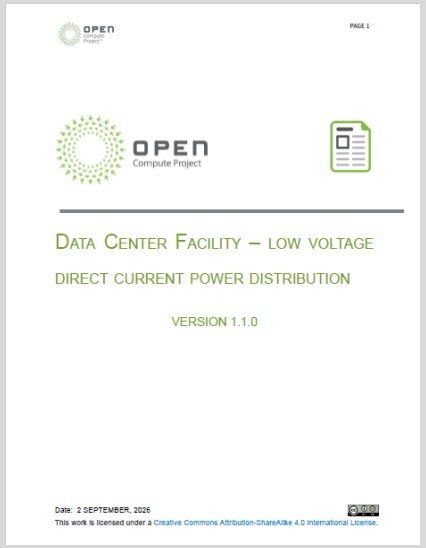
IEC Mapped Standards (2 of 2)

Overcurrent/protection devices			Connectors		
UL 489	**IEC 60947-2**	Molded-case circuit breakers and associated devices	UL 1059, UL 60947-7-1	**IEC 60947-7-1**	Terminal blocks
UL 1066	**IEC 60947-2**	Low-voltage power circuit breakers and assoc. devices	UL 1977	**IEC 61984**	Busbar connectors
UL 489I	**IEC 60947-10**	Solid-state circuit breakers	UL 1691, UL 1682	**IEC 60309 series, IEC TS 63641**	Single-pole, Pin & sleeve connectors
UL 248-4, -8, -10, -12, -15	**IEC 60269 series**	Branch-circuit fuses (Class CC, J, L, R, T)	UL 2237		Multi-point interconnection power cable assemblies
UL 248-13	**IEC 60269 series**	Semiconductor fuses	New OOI		LVDC Litz wire PDU2 & J-box connectors
UL 248-14	**IEC 60127 series**	Supplemental fuses	UL 486A, UL 486B	**IEC 60999 series, IEC 61238 series**	Wire connectors
UL 1449	**IEC 61643 series**	Surge protection devices	UL 1953 OOI	**IEC 60947-7-1**	Power distribution blocks
New OOI		DC AFCI	**Grounding/bonding**		
New OOI	**IEC 60755-1, TS 63053**	DC GFCI	UL 467	**IEC 60364 series**	Grounding/bonding equipment
New OOI	**IEC 61557-8, -9, -15**	Isolation monitors	**Power utilization**		
UL 1053 and New OOI	**IEC 63112?, IEC 61557-19**	Ground-fault sensing and relaying equipment	UL 62368-1	**IEC 62368-1 4th Ed., EN 62368-1 2nd E.**	Compute rack/row equipment (GPUs, shelves, servers)
Cables/cords					
UL 817A OOI		AC/DC power whips	New OOI		Superconducting Cable Systems
UL 2137 OOI		Data center power cables	New OOI		Extruded Insulated Busbars
New OOI		Data center liquid-cooling cables	New OOI		Extruded Liquid Cooled Insulated Busbars
Reference standards					
UL 840	**IEC 60664-1, TS 62993**	Insulation coordination including clearances and creepage distances	UL RP 2986	**TS 63527, EN 50110**	Measuring incident energy exposures
	IEC 60038	IEC Standard Voltages			

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LVDC Power Distribution White Paper

OCP white paper | Data Center Facility - LVDC Power Distribution V1.1



Scan QR code to access white paper

Standards Master Reference File

	A	B	C	D	E	F	G	H	I	J	K	L
1	**Organization**	IEC										
2	**Owner**	Uma Manickam-Chauhan (Siemens), Arnaud Brozek (Socomec)										
3												
4							Do they exist?	Have they been vetted?	Do they exist?	Have they been vetted?		
5	**Product Type**	**Category**	**Product Description**	**Product or System Standard**	**Standard / Code Ref**	**Document Reference (insert link)**	**Requirements for 800VDC (maximum available voltage)**		**Requirements for 1500VDC (maximum available voltage)**		**Action(s) needed to operationalize and deploy?**	**DC Voltage Applicati Limitiation(s)? (if applicable)**
6	Utility Interconnection	Microgrid	guideline for the planning and design DC or hybrid microgrids	System	IEC TS 63354		Yes		Yes			
7	Utility Interconnection	Microgrid	Grid code compliance assessment methods for grid connection of renewable energy	System	IEC TS 63102							
8	Utility Interconnection	Microgrid	LVDC systems - Assessment of standard voltages and power quality	System	IEC TR 63282	IEC TR 63282:2024 | IEC	Yes		Yes			
9	Utility Interconnection	Microgrid	Flexible interconnection systems with LVDC	System	IEC TR 63282-103		Yes		Yes			
10	Electrical Distribution	Other	Low-voltage electrical installations	System	IEC 60364 series	IEC 60364-8-82 | IEC	Yes		Yes			
11	Electrical Distribution	Other	Solar photovoltaic (PV) power supply installations	System	IEC 60364-7-712	IEC 60364-7-712:2025 | IE	Yes		Yes			
12	Electrical Distribution	Other	Prosumer's low-voltage electrical installations	System	IEC 60364-8-82	IEC 60364-8-82:2022 | IEC	Yes		Yes			
13	Energy Storage	BESS - Stationary Applications	Electrical Energy Storage (EES) systems	System	IEC 62933 series also TR and TS		Yes		Yes			
14	Power Conversion	Other	Safety requirements for power electronic converter systems and equipment	Product	IEC 62477-1	IEC 62477-1:2022 | IEC	Yes		Yes			

Full list of all system and product-level standards for 800V DC power distribution architecture
OCP LVDC Power Distribution - Codes & Standards Master Inventory

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Call to Action

- Problems still to solve
  - Closing standard gaps for 800V DC requirements
  - Harmonization/globalization
  - System-level design architecture & validations
  - Industry “Known Unknowns”
    - DC Arc Flash & Worker Protections
    - Capacitive faults in DC circuits
    - Evolving installation codes
    - Utilities and interconnection
- How to get involved
  - Join Codes & Standards workstream
  - Sign up for mailing list from Power Distribution Sub-Project:
    - https://www.opencompute.org/community/power-distribution

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Thank You!



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